

Analysis of agricultural drought using vegetation temperature condition index (VTCI) from Terra/MODIS satellite data

N. R. Patel · B. R. Parida · V. Venus · S. K. Saha ·
V. K. Dadhwal

Received: 28 January 2009 / Accepted: 8 December 2011 / Published online: 27 December 2011
© Springer Science+Business Media B.V. 2011

Abstract The most commonly used normalized difference vegetation index (NDVI) from remote sensing often fall short in real-time drought monitoring due to a lagged vegetation response to drought. Therefore, research recently emphasized on the use of combination of surface temperature and NDVI which provides vegetation and moisture conditions simultaneously. Since drought stress effects on agriculture are closely linked

to actual evapotranspiration, we used a vegetation temperature condition index (VTCI) which is more closely related to crop water status and holds a key place in real-time drought monitoring and assessment. In this study, NDVI and land surface temperature (T_s) from MODIS 8-day composite data during cloud-free period (September–October) were adopted to construct an NDVI– T_s space, from which the VTCI was computed. The crop moisture index (based on estimates of potential evapotranspiration and soil moisture depletion) was calculated to represent soil moisture stress on weekly basis for 20 weather monitoring stations. Correlation and regression analysis were attempted to relate VTCI with crop moisture status and crop performance. VTCI was found to accurately access the degree and spatial extent of drought stress in all years (2000, 2002, and 2004). The temporal variation of VTCI also provides drought pattern changes over space and time. Results showed significant and positive relations between CMI (crop moisture index) and VTCI observed particularly during prominent drought periods which proved VTCI as an ideal index to monitor terminal drought at regional scale. VTCI had significant positive relationship with yield but weakly related to crop anomalies. Duration of terminal drought stress derived from VTCI has a significant negative relationship with yields of major grain and oilseeds crops, particularly, groundnut.

N. R. Patel (✉) · S. K. Saha
Indian Institute of Remote Sensing (ISRO),
4-Kalidas Road,
Dehradun 248001 Uttanchal, India
e-mail: nrpatel@iirs.gov.in

S. K. Saha
e-mail: sksaha@iirs.gov.in

B. R. Parida
Max-Planck Institute of Meteorology,
Hamburg, Germany
B. R. Parida
e-mail: bikashrp@gmail.com

V. Venus
Department of Natural Resources, International Institute
for Geo-Information Science and Earth Observation (ITC),
Enschede, The Netherlands
e-mail: venus@itc.nl

V. K. Dadhwal
National Remote Sensing Centre (ISRO),
Balanagar, Hyderabad 500 625, India
e-mail: dadhwalvk@gmail.com

Keywords Remote sensing · MODIS · T_s -NDVI space · VTCI · Drought · Crop yields

Introduction

Drought produces a complex web of impacts, which spans many sectors of the economy, especially the agriculture sector. Drought leads to decline in food grain production depending upon the intensity, duration, and spatial coverage of drought stress. Over 68% of India is vulnerable to drought. Large parts of the country perennially reel under recurring drought. The drought-prone areas of the country are mainly confined to western and peninsular India—primarily arid, semi-arid, and sub-humid regions. Recently in 2002–2003, India has faced one of the worst and unique drought episode in terms of magnitude, spacing, dispersion, and duration. Drought in India has far ranging impacts on all natural hazards from human and livestock death toll to economical losses. Thus, monitoring of the most disastrous phenomena of drought is of utmost concern to planner or decision makers from view point of food security and trade as a whole. In India, the point-based meteorological drought indices have been extensively used for drought monitoring. But sparse meteorological network and lack of timely availability of weather data always hinders the accurate and timely monitoring of regional drought.

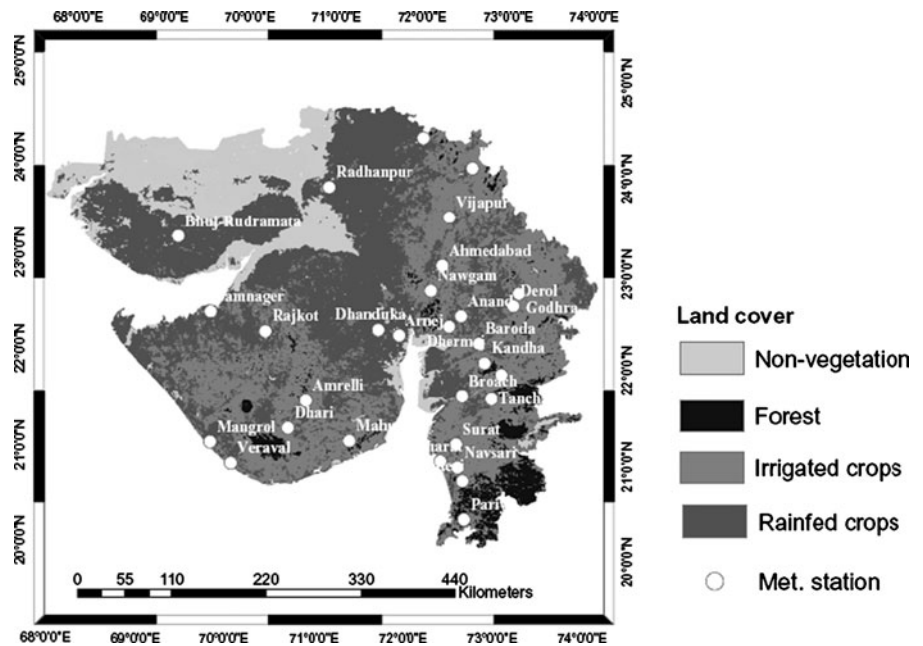
Innovations in remote sensing technology have provided newer dimensions of spatial solutions to many environmental problems, including natural hazard monitoring. Satellite remote sensing has become crucial particularly for timely detection and monitoring of drought due to more prompt availability of spatiotemporal data over entire globe. The most commonly used normalized difference vegetation index (NDVI) from remote sensing often fall short in real-time drought monitoring (Park et al. 2004; Wang et al. 2004). Due to a lagged vegetation response to drought, NDVI cannot detect drought events instantaneously. On the other hand, surface temperature (T_s) is sensitive to water stress and has been identified as good indicator of water stress (Jackson et al. 1981; Goetz 1997). Thus, accurate and real-time drought monitoring needs combination of the thermal and visible/near-infrared wavelength to provide information on vegetation and moisture condition simultaneously. The scatter plot of remotely sensed temperature and spectral vegetation index often exhibits a triangular (Carlson

et al. 1994) or trapezoidal (Moran et al. 1994) shape and is called the Ts-NDVI space if a full range of fractional vegetation cover and soil moisture content is represented. Recently, this Ts-NDVI space has been widely exploited to derive various types of hydrological information such as air temperature (Boegh et al. 1998), evapotranspiration (Carlson et al. 1995), and soil moisture (Moran et al. 1994; Gillies et al. 1997; Sandholt et al. 2002). Drought stress effects on agriculture are closely linked to actual evapotranspiration by crop canopies throughout the growth period. Therefore, drought index which is closely related to crop water status holds a key place in drought monitoring. Recently, many drought indices like temperature vegetation dryness index, vegetation temperature condition index (VTCI), water deficit index, and the crop water stress index have been defined for quantification and real-time monitoring of the spatial extent and magnitude of drought (Nemani et al. 1993; Moran et al. 1994; Wan et al. 2004). Hence, looking into enormous potential of VIT trapezoid indices, the present study was undertaken to investigate the potential of vegetation/temperature condition index from Terra/MODIS satellites for real-time assessment of agricultural drought in semi-arid regions of India.

Description of the study area

The study was carried out in Gujarat State of India situated between 20°01' to 24°07' N latitudes and 68°04' to 74°04' E longitudes. The topography includes a central high land in the northeastern part, western hills in the southeastern part and west coast in the central part (comprising Gujarat plain, Kathiawar Peninsula, and Kachchh Peninsula). The climate represents a wide variability ranging from arid, through semi-arid, to sub-humid tropical monsoonal type. The arid and semi-arid areas of the state occupy 4.9% and 33.6%, respectively. The annual normal rainfall varies from about 400 mm at northwest end of the state to about 2,500 mm at southeast end of the state. The amount of rain and its distribution is highly erratic over time and space, particularly, in the arid and semi-arid areas of the state. The length of growing period ranges from 90 to 150 days. In general,

Fig. 1 Location of meteorological stations and land cover in study region



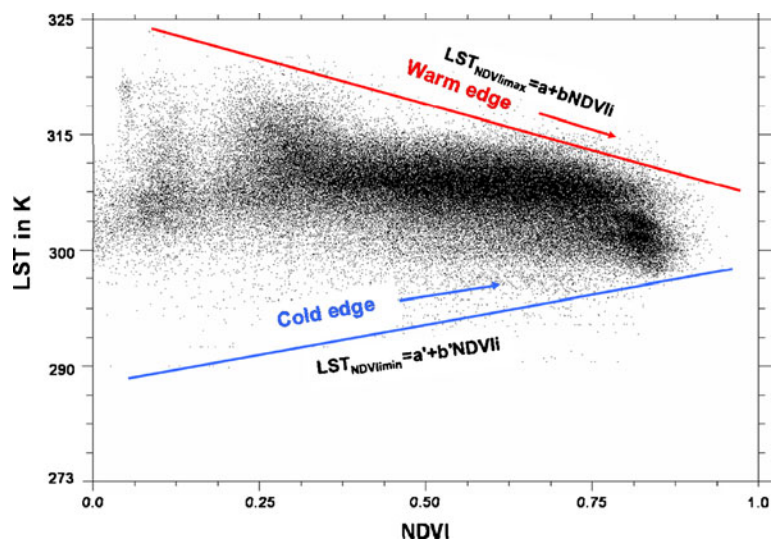
districts located in north, northeast, and northwest parts of the state suffer from drought or water scarcity with a re-occurrence interval of 3–4 years. The major land use is agriculture (50%). Of the total cropped area, food crops like cereals and pulses account for about 50%, while the remaining area is oilseed, fiber, and fodder crops. The forest area is 10% and distributed all along the eastern border and hilly parts of Kathiawar Peninsula.

Materials and methods

Datasets

Terra/MODIS land products were downloaded from EOS data gateway (<http://edcimswww.cr.usgs.gov/pub/imswelcome/>) during a period ranges from 241 to 297 days of year (DOY) in each of 5 years (2000–2004). The satellite data were (1) Terra/MODIS Land

Fig. 2 Parameterization of $LST_{NDVI_{max}}$ and $LST_{NDVI_{min}}$ from $LST/NDVI$ triangle



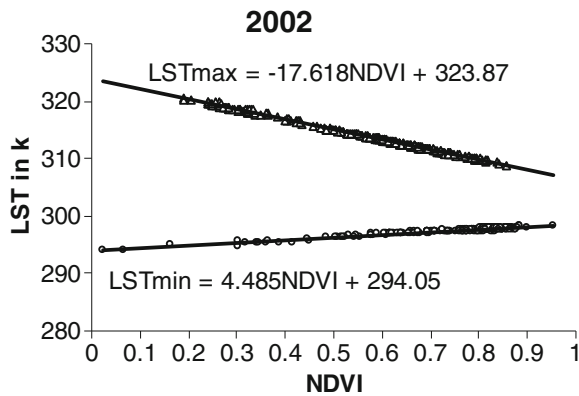


Fig. 3 Linear equation showing LSTmax and LSTmin from The Ts-NDVI space on DOY 257 in 2002

Surface Temperature/Emissivity 8-Day L3 Global 1 km SIN Grid V004 (MOD11A2) and (2) Terra/MODIS Surface Reflectance 8-Day L3 Global 250 m SIN Grid V004 (MOD09Q1). These data are distributed by the

Land Processes Distributed Active Archive Center (LP DAAC) located at the US Geological Survey (USGS) Center for Earth Resources Observation and Science (EROS) (<http://LPDAAC.usgs.gov>). These data were re-projected to Albers equal area projection.

The historical (25 years) daily meteorological data for 30 meteorological stations were obtained from the Department of Agrometeorology (GAU, Anand) and India Meteorological Department. These stations were uniformly distributed over the entire Gujarat State (See Fig. 1). The long-term meteorological data include maximum temperature, minimum temperature, and rainfall. Available soil water capacity of each station was obtained from soil-climatic databases generated by National Bureau of Soil Survey and Land Use Planning, India. The historical crop statistics on cereals (rice, pearl millet, sorghum, and other coarse millets), pulses (chickpea, mung, pigeonpea, blackgram, etc.) and oilseeds grown during kharif (June–

Table 1 The warm and cold edges in NDVI-Ts space estimated by linear regression for drought (2000 and 2002) and wet (2004) years

Year	DOY	Warm edge ($T_{\max}$)	R^2	Cold edge ($T_{\min}$)	R^2
2000	241	$LST_{NDVI_{\max}} = -14.581NDVI + 319.63$	0.99	$LST_{NDVI_{\min}} = 9.9859NDVI + 286.28$	0.99
	249	$LST_{NDVI_{\max}} = -15.992NDVI + 328.46$	0.95	$LST_{NDVI_{\min}} = 4.3434NDVI + 292.82$	0.86
	257	$LST_{NDVI_{\max}} = -16.211NDVI + 328.23$	0.99	$LST_{NDVI_{\min}} = 1.921NDVI + 298.69$	0.94
	265	$LST_{NDVI_{\max}} = -14.805NDVI + 329.08$	0.99	$LST_{NDVI_{\min}} = 3.3155NDVI + 300$	0.97
	273	$LST_{NDVI_{\max}} = -23.37NDVI + 331.86$	0.99	$LST_{NDVI_{\min}} = 0.8475NDVI + 300.94$	0.60
	281	$LST_{NDVI_{\max}} = -22.235NDVI + 330.79$	0.99	$LST_{NDVI_{\min}} = 3.5288NDVI + 297.22$	0.95
	289	$LST_{NDVI_{\max}} = -21.877NDVI + 330.88$	0.99	$LST_{NDVI_{\min}} = 1.3253NDVI + 300.86$	0.90
	297	$LST_{NDVI_{\max}} = -23.193NDVI + 328.17$	0.99	$LST_{NDVI_{\min}} = 2.8904NDVI + 301.84$	0.99
2002	241	$LST_{NDVI_{\max}} = -12.499NDVI + 309.79$	0.99	$LST_{NDVI_{\min}} = 9.3349NDVI + 286.82$	0.95
	249	$LST_{NDVI_{\max}} = -13.209NDVI + 320.62$	0.99	$LST_{NDVI_{\min}} = 5.3852NDVI + 292.75$	0.96
	257	$LST_{NDVI_{\max}} = -17.618NDVI + 323.87$	0.99	$LST_{NDVI_{\min}} = 4.485NDVI + 294.05$	0.94
	265	$LST_{NDVI_{\max}} = -19.581NDVI + 328.03$	0.99	$LST_{NDVI_{\min}} = 2.232NDVI + 298.21$	0.90
	273	$LST_{NDVI_{\max}} = -15.258NDVI + 327.51$	0.99	$LST_{NDVI_{\min}} = 3.8621NDVI + 301.62$	0.98
	281	$LST_{NDVI_{\max}} = -16.874NDVI + 326.94$	0.99	$LST_{NDVI_{\min}} = 4.8594NDVI + 300.67$	0.97
	289	$LST_{NDVI_{\max}} = -15.613NDVI + 324.84$	0.99	$LST_{NDVI_{\min}} = 3.1049NDVI + 300.32$	0.96
	297	$LST_{NDVI_{\max}} = -17.761NDVI + 325.38$	0.99	$LST_{NDVI_{\min}} = 3.4135NDVI + 301.21$	0.98
2004	241	$LST_{NDVI_{\max}} = -21.066NDVI + 324.03$	0.95	$LST_{NDVI_{\min}} = 8.7953NDVI + 289.29$	0.99
	249	$LST_{NDVI_{\max}} = -17.951NDVI + 323.04$	0.99	$LST_{NDVI_{\min}} = 7.6205NDVI + 292.4$	0.94
	257	$LST_{NDVI_{\max}} = -17.493NDVI + 322.26$	0.99	$LST_{NDVI_{\min}} = 8.6475NDVI + 290.76$	0.97
	265	$LST_{NDVI_{\max}} = -20.737NDVI + 323.93$	0.99	$LST_{NDVI_{\min}} = 5.7734NDVI + 292.95$	0.91
	273	$LST_{NDVI_{\max}} = -25.74NDVI + 327.9$	0.97	$LST_{NDVI_{\min}} = 13.089NDVI + 285$	0.98
	281	$LST_{NDVI_{\max}} = -15.333NDVI + 319.14$	0.97	$LST_{NDVI_{\min}} = 3.2698NDVI + 295.59$	0.89
	289	$LST_{NDVI_{\max}} = -14.615NDVI + 322.03$	0.99	$LST_{NDVI_{\min}} = 5.9779NDVI + 299.33$	0.98
	297	$LST_{NDVI_{\max}} = -19.104NDVI + 323.18$	0.99	$LST_{NDVI_{\min}} = 11.341NDVI + 296.11$	0.98

October) were obtained from the Department of Agriculture, government of Gujarat. These statistics were used to compute long-term (1981–2000) time series of total food grains and oilseeds productivity during kharif season for each district in Gujarat.

Vegetation temperature condition index

This index has been used as an effective agricultural drought index to monitor the spatial pattern of vegetation over a region. The VTCI approach (Wan et al. 2004) for agricultural drought monitoring integrates remotely sensed land surface reflectance and thermal properties, and gives emphasis on changes in both land surface temperature and NDVI based on Ts-NDVI space relation. Mathematically, it can be written as

$$VTCI = \frac{LST_{NDVI_{limax}} - LST_{NDVI_i}}{LST_{NDVI_{limax}} - LST_{NDVI_{limin}}} \quad (1)$$

Where

$$\begin{aligned} LST_{NDVI_{limax}} &= a + b \text{ NDVI}_i \\ LST_{NDVI_{limin}} &= a' + b' \text{ NDVI}_i \end{aligned} \quad (2)$$

The $LST_{NDVI_{limax}}$ and $LST_{NDVI_{limin}}$ are maximum and minimum LSTs of pixels, which have the same NDVI value in a study region on each DOY or period of image, respectively. LST_{NDVI_i} denotes actual LST of one pixel whose NDVI value is $NDVI_i$. Coefficients a , b , a' , and b' can be estimated from an area large enough where soil moisture at surface layer should span from wilting point to field capacity at pixel level. The shape of the scatter plot is normally triangular at a regional scale (Gillies et al. 1997; Wang et al. 2004) if the study area is large enough to provide a wide range of NDVI and surface moisture conditions. In the present study, all 8-day images of LST and NDVI covering area of 196,000 km² were processed to correctly obtained VTCI over study area.

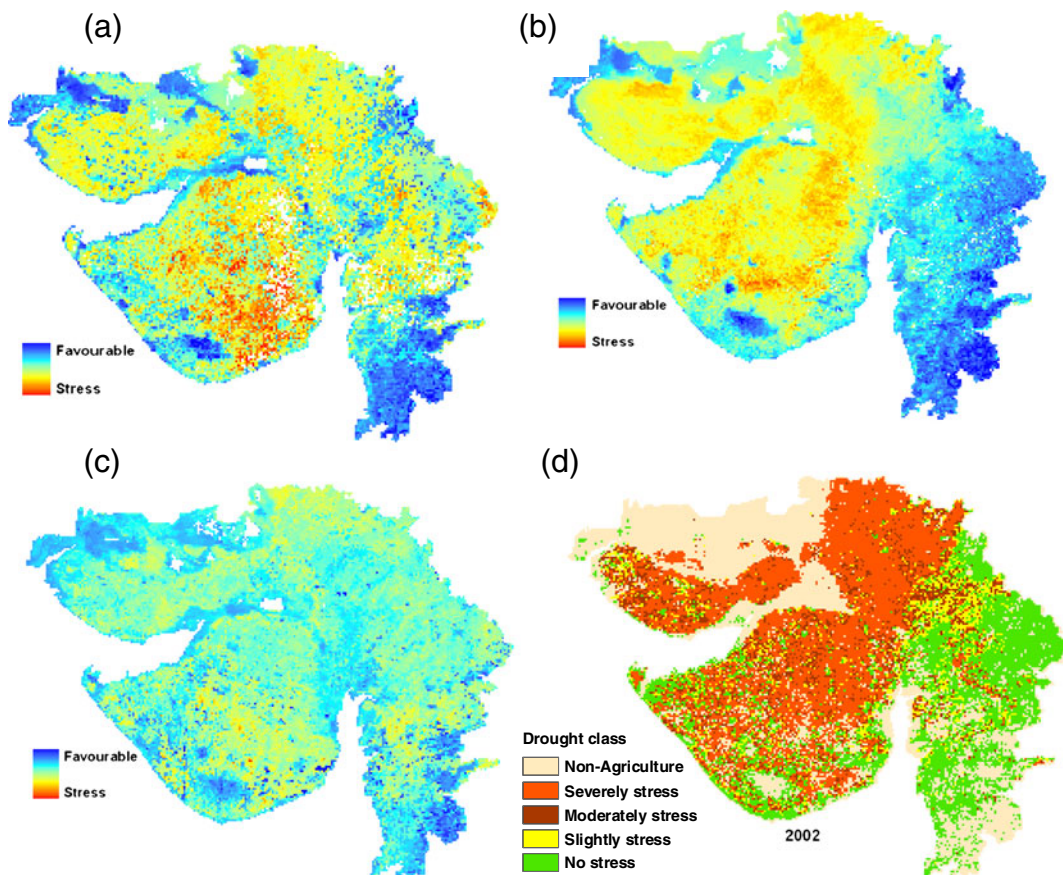


Fig. 4 Spatial pattern of VTCI on DOY 249 in years **a** 2000, **b** 2002, **c** 2004, and **d** drought severity (2002)

VTCI is based on the interpretation of a simplified NDVI-Ts space (Fig. 2), where $LST_{NDVI_{max}}$ and $LST_{NDVI_{min}}$ represent “warm edge” (water stress restriction) and “cold edge” (no water limitation). The 2D scatter plot of NDVI and LST data has been made using ENVI image processing software. Isolines, i.e., warm edge and cold edge, were extracted from 2D plot by sorting both maximum and minimum surface temperature for each small interval of NDVI (0.01). The coefficients a , b , a' , and b' can be derived for these edges by fitting least square regression procedure. In Eq. 2, a and a' are the intercept and b and b' are the slope of the warm edge and cold edge, respectively. Based on the obtained NDVI-Ts space, the VTCI was calculated for each pixel, excluding water body pixels using band math in ENVI. The value of VTCI ranges from 0 to 1. In general, lower values of VTCI indicate higher magnitudes of drought stress.

Crop moisture index

The crop moisture index (CMI) developed by Palmer (1968) is a complement to the Palmer Drought Severity Index (PDSI). It measures the degree to which crop moisture requirements are met and is more responsive to short-term changes in moisture conditions and is not intended to assess long-term droughts. CMI is normally calculated with a weekly time step and is primarily based on the mean temperature, total precipitation for each week, and the CMI value from the previous week. Each growing season, CMI typically begins and ends near zero. CMI gives the short term or current status of purely agricultural drought or moisture surplus and can change rapidly from week to week. The CMI for each meteorological station has been computed using PDSI program developed by University of Nebraska, Lincoln, USA (<http://nadss.unl.edu>). The weekly time series of historical (1981–2005) precipitation and mean temperature along with available water capacity were used as inputs.

De-trending of long-term crop productivity

The time series of food grain and oilseeds productivity were linearly de-trended before correlations were calculated to remove the effect of improvements in crop science, technology, etc. The de-trended yields were

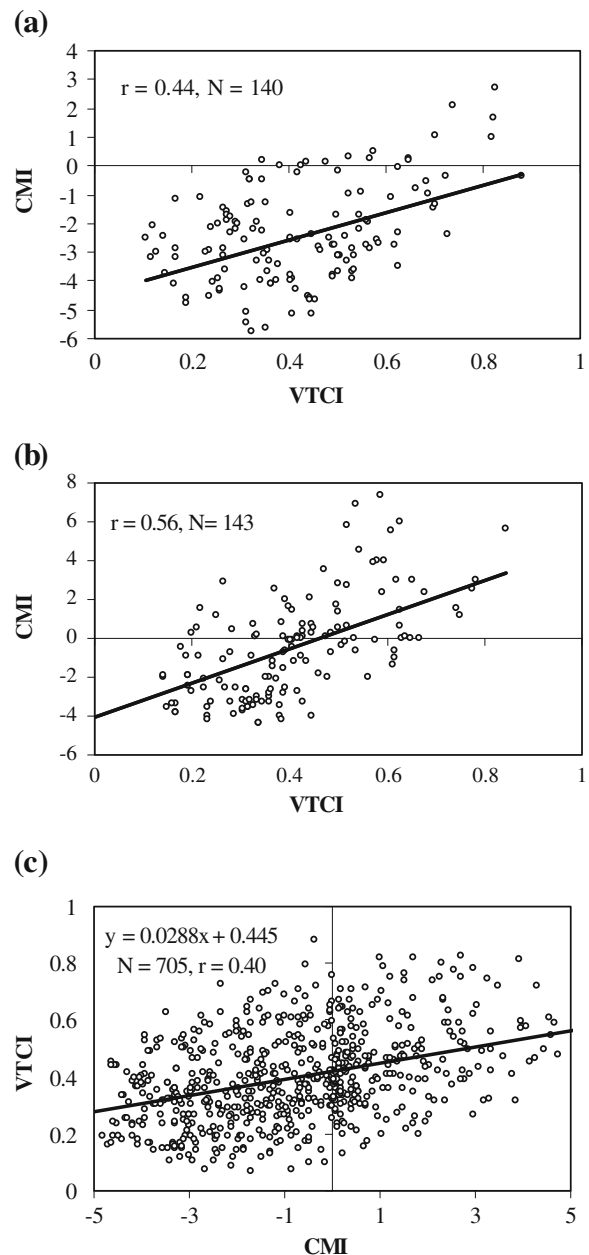


Fig. 5 Relationship between CMI and VTCI during 2000 and 2002 and pooled data of 5 years

obtained by subtracting the per year yield variation from historical record of reported yields of food grain and oilseeds (Larson et al. 2004). The de-trended yield anomaly (DYa) is computed as:

$$DYa_i = [((Y_{a_i})/(Y_{t_i}) - 1)100] \quad (3)$$

Where, DYa_i is de-trended food grain yield anomaly for the i th year, Y_{a_i} and Y_{t_i} are actual and time trend based

yield ($Y_t = a + b \times \text{year}$) of i th year, respectively. The slope (b) of this regression line for each district was used as an indicator of the overall trend in productivity of food grain and oilseeds.

Results and discussions

LST-NDVI space

The LST-NDVI isolines for every 8-day composite interval from DOY 241 to 297 were formed by extracting maximum and minimum land surface from Ts-NDVI scatter plot of whole study area by using small interval of NDVI. For example, isolines of LST_{max} and LST_{min} obtained from Ts-NDVI space for DOY 257 in 2001 is shown in Fig. 3. Similarly, warm and cold edges for every 8-day composite images of LST and NDVI were defined to calculate VTCI for each 8-day interval for time during drought (2000 and 2002) and normal (2004) years. The warm and cold edges are presented only for drought years (Table 1). The results presented in Table 1 reveal that the slope of warm edge was negative for every interval during both years, whereas the slope for cold edge was consistently positive. The slope of warm edge indicates the T_{max} decreasing as NDVI increases for every NDVI interval. The positive slope of cold edge in all Ts-NDVI spaces indicates the LST_{min} increasing as NDVI increases. These results are in agreement with those reported by Wang et al. (2004). In general for warm edge and cold edge, it has been observed that the intercept value is higher during drought years 2000, 2001, and 2002 as

compared to non-drought year 2003. The overall coefficient of determination (R^2) varies from 0.90 to 0.99 in NDVI-Ts space.

Spatial pattern of VTCI

The spatial patterns of VTCI on DOY 249 of both drought (2000 and 2002) and normal (2004) years are shown in Fig. 4. The lower VTCI values denote the stress condition while higher VTCI values denote favorable conditions in the region. On DOY 249, drought years, i.e., 2000 and 2002 had lower VTCI values across study area as against those of 2004. The VTCI map has been stretched into lower VTCI values to higher VTCI values and it corresponds to stress conditions respectively. During 2004, VTCI values were more than 0.5 due to favorable conditions.

In general, Northern Gujarat, Central Gujarat, and Saurashtra regions are showing stress conditions during 2000 and 2002. These areas showed lower VTCI values as compared to other area. A few districts like Vadodara, Bharuch, and Normada in South Gujarat also showed lower VTCI values. Other southern districts had higher VTCI values ($\text{VTCI} > 0.75$) and implied that these districts were less affected by drought stress. The higher VTCI values occurred in some areas even during drought years was mainly ascribed to the availability of irrigation facilities which tends to reduce moisture stress to the agricultural crops. In addition to this, the few districts in extreme south are covered by forest. However, on same DOY 249 in 2004, the entire study area represents a balanced VTCI having value of more than 0.45.

Fig. 6 Temporal variation in percentage of agricultural area affected by drought stress

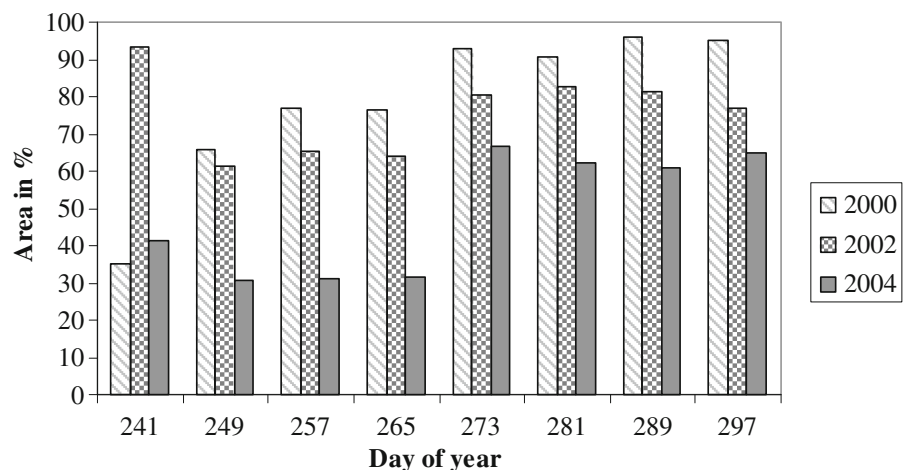


Table 2 Correlation coefficient of VTCI with yield and de-trended yield anomalies

DOY	Food grains				Oilseeds	
	2000		2002		2000	2002
	Yield	Anomaly	Yield	Anomaly	Anomaly	Anomaly
241	0.414 ^a	−0.036	0.46 ^a	0.342	0.033	0.206
249	0.589 ^a	0.233	0.54 ^a	0.451 ^a	−0.079	0.275
257	0.542 ^a	0.273	0.49 ^a	0.330	−0.087	0.221
265	0.451 ^a	0.105	0.38	0.189	−0.044	0.237
273	0.559 ^a	0.186	0.42 ^a	0.192	−0.075	0.315
281	0.436 ^a	0.138	0.39	0.212	−0.043	0.282
289	0.509 ^a	0.150	0.30	0.166	−0.025	0.319
297	0.464 ^a	0.214	0.39	0.241	0.114	0.331

^a Significant at 95% level (p value<0.05) N=25 districts

Relations of VTCI with crop moisture index

Spatial patterns of VTCI at different periods during drought and wet years were compared with corresponding weekly crop moisture index calculated at the meteorological stations. These relations show an ability of VTCI to detect the magnitude of short-term agricultural drought. Results depicted in Fig. 5 show that VTCI had a significantly positive relation with crop moisture index, particularly during drought years (2000 and 2002). The higher R^2 (0.31) for VTCI/CMI relation observed during 2002 than 2000 ($R^2=0.19$) was mainly attributed to high degree of variability in crop moisture index representing dry and wet conditions experienced during 2002. These relations also illustrate that the value of VTCI increased with an increase in crop moisture availability in the crop root zone. Furthermore, a

relationship of VTCI against CMI constructed using all observation of 5 years reveals that VTCI values below 0.45 corresponds to negative CMI which represents drought stress conditions. These relations help to identify thresholds to categorized spatial patterns of VTCI corresponding to different classes of drought severity based on crop moisture index. For example, the VTCI on DOY 249 in year 2002 was categorized into normal (0.45–1.0), slightly stress (0.45–0.38), moderately stress (0.38–0.31), and severely stress (0.31–0.0).

Temporal variation in spatial extent of drought affected area

All VTCI images at 8-day intervals during drought (2000 and 2002) and normal (2004) years were categorized into drought (VTCI<0.45) and no-drought

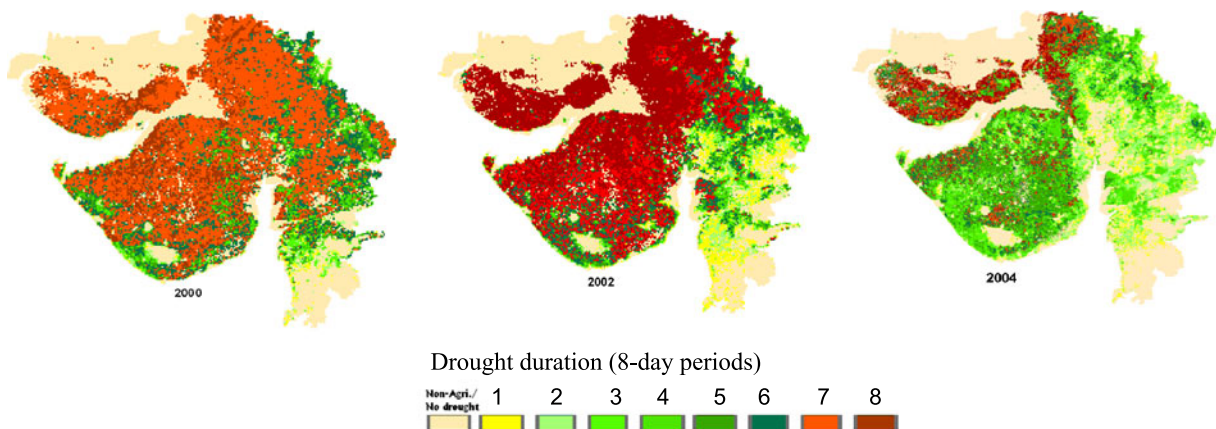


Fig. 7 Spatial pattern of drought duration obtained from VTCI for drought (2000 and 2002) and wet (2004) years

($VTIC > 0.45$) based on $VTIC$ threshold and pixels corresponds to agriculture were aggregated for each drought class for every interval. The resultant agricultural areas affected by drought at each 8-day intervals are shown in Fig. 6. Results indicated that agricultural area affected by drought stress was consistently higher in 2002. The magnitude of percentage area affected stood greater than 50 for all DOY except DOY 241. The area affected by drought in 2002 also had good agreement with seasonally 57% agricultural drought affected area reported by Planning Commission, Government of India (<http://planningcommission.nic.in>). In 2002, the drought stress area is greater on DOY 241 as compared to other days of year (249–297). This indicates that the year 2002 had faced an early drought. During 2004, the agricultural area under stress is less as compared to 2000 and 2002. The drought affected area was found less during initial periods up to DOY 265 in 2004 but suddenly increased from DOY 273 to 297. In general, it has been observed that 297 JD represents a greater area under stress for all the year. Although the year 2004 was not a drought year, still stress is indicated. This is due to the senescence of most of the crops during 297 JD. The DOY 257 represented a better result which describes the different drought class area in percentage. It indicates that most of the agriculture area has been affected by drought during 2000. Area affected by drought in 2002 was lesser than 2000 and have correspondence with conventional drought reports. The year 2004 appeared to be normal as evident from least percentage of area affected by drought stress.

Drought stress and its effects on crop performance

It is well understood that $VTIC$ allows the detection and monitoring of the spatial extent of drought-affected agricultural area. The effect of drought stress on crop performance was also analyzed by comparing district-wise average $VTIC$ at different time interval with linearly de-trended yields of food grains and oilseeds at district level. The de-trended yield anomaly of food grains and oilseeds were negative for the majority of districts during both 2000 and 2002 drought years. The average yield anomalies of food grains over districts were -28.17% and -11.20% for years 2000 and 2002, respectively. Similarly, the yield anomalies of oilseeds were -20.5% and -35.5% for years 2000 and 2002, respectively. These negative anomalies reveal a drastic

reduction in crop productivity due to drought stress in 2000 and 2002. We found significant and positive association between $VTIC$ and productivity of food grains and yields during all DOY. Correlation between $VTIC$ and yield anomalies exhibits a weak but positive correlation in all periods (Table 2). These positive correlations indicate that $VTIC$ has ability to detect drought stress causing any reduction in productivity of food grains and oilseeds. The high magnitude of correlation coefficient on DOY 249 and 257 reveals more sensitivity of these periods to drought stress.

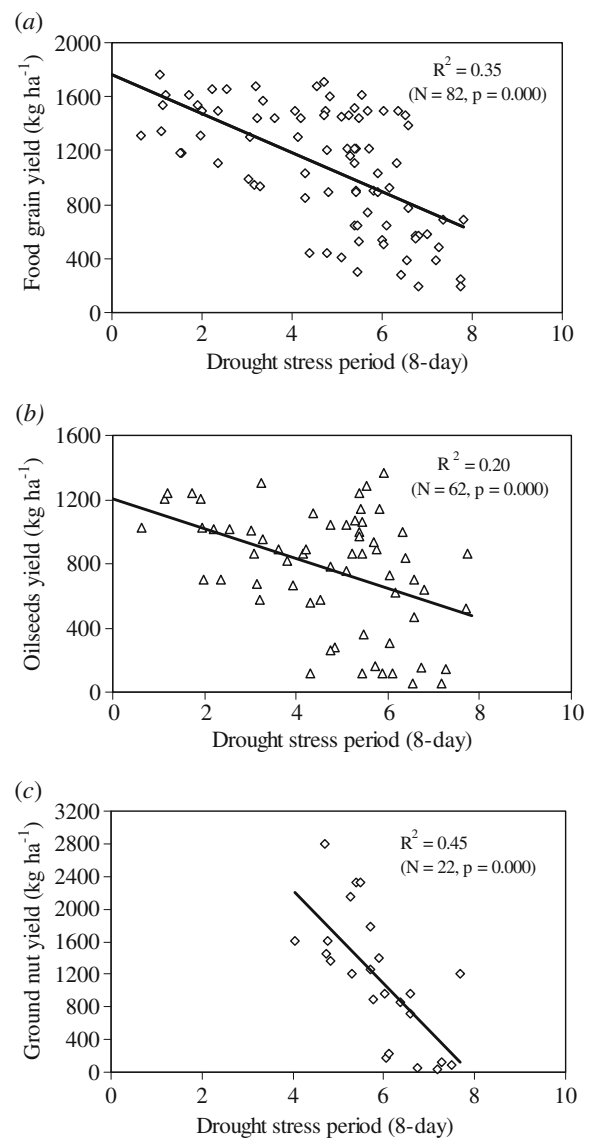


Fig. 8 Comparison of drought duration with yields of **a** food grains, **b** oilseeds, and **c** groundnut

Drought duration from VTCI and crop yield

All VTCI images at 8-day intervals during drought (2000 and 2002) and normal (2004) were categorized into drought/no drought based on VTCI thresholds. The threshold value of 0.45 was used generate a binary image at each 8-day composite intervals showing drought ($VTCI \leq 0.45$) and no-drought class ($VTCI > 0.45$). All binary images were summed to obtained drought duration which represents as number of 8-day period with drought/no drought condition. The spatial patterns of drought duration during reproductive period (i.e., DOY 241–297), as shown in Fig. 7, clearly depicts that a majority of the areas in the Gujarat State have experienced drought stress during seven periods (8-day) in year 2000. Whereas, year 2004 was least affected and has experienced only three to four periods of drought stress. These three to four periods of drought stress observed during normal year were actually due to natural senescence of crop canopies occurring during crop's maturity. However, year 2002 had a majority of the areas under drought during all periods of reproductive growth. It could also be inferred that drought effects during 2000 and 2002 were severe particularly in Kathiawar and Kachchh Peninsula characterized by high intensity and duration of drought.

The significance of drought duration obtained from VTCI images was further validated by comparing it with actual crop's performance. The relations between district level zonal mean of drought duration and yields of food grains and oilseeds (Fig. 8a, b) indicate that yields of food grains and oilseeds declined significantly with increase in duration of drought. The observed relations were significant at 99% probability ($p < 0.01$). Drought duration was found more closely related to food grain yield than to oilseeds. Significant but weak relationship ($R^2 = 0.20$) between drought duration and oilseed yield was mainly attributed to relatively less dominance of oilseed crops in districts other than those falls within Kathiawar Peninsula. However, when drought duration was compared to yields of groundnut a pre-dominantly cultivated oilseed crop in Kathiawar Peninsula, the relation was found highly significant (Fig. 8c).

Conclusions

This study explores integration of vegetation represented by NDVI and surface temperature measurements

for deriving vegetation temperature condition index (VTCI) based on interpretation of TS-NDVI space. In this study, the role of satellite-derived VTCI for monitoring and assessing agricultural drought on a regional scale has been shown by integrating meteorologically based CMI. It was observed that the VTCI is an ideal index to monitor agricultural drought because it is highly sensitive to short-term changes in crop moisture status in terms of CMI. VTCI was also found to be related crop yield anomalies occurring due to drought conditions. Furthermore, interpretation of drought duration from time series of VTCI can clearly depicts severity of drought stress effects on crop performance. The major advantage of VTCI is that it captures information about drought stress condition solely by satellite measurements. However, this index has limited applicability in presence of cloudy condition since surface temperature is highly sensitive to cloud cover.

Acknowledgments Authors would like to thank the MODIS Land Discipline Group for creating and sharing the MODIS LAND data. We also express our sincere thanks to Department of Agrometeorology, AAU, Anand (India) and Department of Agriculture, Gandhinagar for sharing meteorological and crop yield statistics.

References

- Boegh, E., Soegaard, H., Hanan, N., Kabat, P., & Lesch, L. (1998). A remote sensing study of the NDVI-Ts relationship and the transpiration from sparse vegetation in the Sahel based on high-resolution satellite data. *Remote Sensing of Environment*, 69, 224–240.
- Carlson, T. N., Gillies, R. R., & Perry, E. M. (1994). A method to make use of thermal infrared temperature and NDVI measurements to infer soil water content and fractional vegetation cover. *Remote Sensing Reviews*, 9, 16–173.
- Carlson, T. N., Gillies, R. R., & Schmugge, T. J. (1995). An interpretation of methodologies for indirect measurement of soil water content. *Agricultural and Forest Meteorology*, 77, 191–205.
- Gillies, R. R., Carlson, T. N., Cui, J., Kustas, W. P., & Humes, K. S. (1997). A verification of the 'triangle' method for obtaining surface soil water content and energy fluxes from remote measurement of the Normalized Difference Vegetation Index (NDVI) and surface radiant temperature. *International Journal of Remote Sensing*, 18, 3145–3166.
- Goetz, S. J. (1997). Multi-sensor analysis of NDVI, surface temperature and biophysical variables at a mixed grassland site. *International Journal of Remote Sensing*, 18, 71–94.
- Jackson, R. D., Idso, S. B., Beginato, R. J., & Pinter, P. J., Jr. (1981). Canopy temperature as a crop water stress indicator. *Water Resources Research*, 17, 1133–1138.

- Larson, D. W., Jones, E., Pannu, R. S., & Sheokand, R. S. (2004). Instability in Indian agriculture-a challenge to the Green Revolution technology. *Food Policy*, 29, 257–273.
- Moran, M. S., Clarke, T. R., Inoue, U., & Vidal, A. (1994). Estimating crop water deficit using the relation between surface air temperature and spectral vegetation index. *Remote Sensing of Environment*, 49, 246–263.
- Nemani, R. R., Pierce, L. L., Running, S. W., & Goward, S. N. (1993). Developing satellite derived estimates of surface moisture status. *Journal of Applied Meteorology*, 28, 276–284.
- Palmer, W. C. (1968). Keeping track of crop moisture conditions, nationwide: The new Crop Moisture Index. *Weatherwise*, 21, 156–161.
- Park, S., Feddema, J. J., & Egberts, S. L. (2004). MODIS land surface temperature composite data and their relationships with climatic water budget factors in the central Great Plains. *International Journal of Remote Sensing*, 26, 1127–1144.
- Sandholt, I., Rasmussen, K., & Anderson, J. (2002). A simple interpretation of the surface temperature/vegetation index space for assessment of the surface moisture status. *Remote Sensing of Environment*, 79, 213–224.
- Wan, Z., Wang, P., & Li, X. (2004). Using MODIS Land Surface Temperature and Normalized Difference Vegetation Index products for monitoring drought in the southern Great Plains, USA. *International Journal of Remote Sensing*, 25, 61–72.
- Wang, C., Qi, S., Niu, Z., & Wang, J. (2004). Evaluating soil moisture status in China using the temperature-vegetation dryness index (TVDI). *Canadian Journal of Remote Sensing*, 30, 671–679.